
Beyond Surface Fidelity: Transformation-Aware Verification for RAG and Stateless Cognitive Adaptation for Educational AI

RESEARCH PROPOSAL

AUTHOR(S):

Ahmad Raza

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
THAL UNIVERSITY BHAKKAR

PAPER 1: TAV (Beyond Surface Fidelity: Transformation-Aware Verification for RAG)

1. Problem Statement

Every existing RAG verification framework i.e., FActScore (Min et al., 2023), VeriScore (Song et al., 2024), VeriCite (Qian et al., 2025), BERTScore (Zhang et al., 2020), NLI entailment (Honovich et al., 2022) operates under one implicit assumption:

The Fidelity Assumption: A faithful generation should approximate the surface form of its source.

This assumption is valid for general-purpose RAG, where the goal is faithful reproduction. It breaks catastrophically for any downstream task requiring semantic transformation (content that must be *entailed by* the source but cannot *resemble* it). We call this the **DERIVED Problem**.

The DERIVED Problem is not domain-specific. It arises wherever faithful generation requires surface-form divergence:

Domain	Source	Faithful Output	Surface Similarity
Educational explanation	" $O(\log n)$ worst-case comparisons"	"About 20 comparisons for a million elements"	Near zero
Biomedical simplification	"Myocardial infarction with ST-elevation"	"A type of heart attack where blood flow is completely blocked"	Near zero
Legal summarization	"The party of the first part shall indemnify..."	"You are responsible if something goes wrong"	Near zero
Scientific journalism	" $p < 0.05$, Cohen's $d = 0.8$ "	"The effect was statistically significant and practically meaningful"	Near zero
Code documentation	$O(n \log n)$	"Scales well — sorting a million items takes roughly 20 million steps"	Near zero

In every case, every existing verifier flags the faithful output as a hallucination. This is not a calibration problem. We prove it is a *fundamental impossibility* of any surface-form verification approach.

2. Core Research Question

Can a derivation-aware verification framework distinguish valid semantic transformations from hallucinations reducing false rejection rate by $\geq 40\%$ relative to surface-matching baselines while maintaining $\geq 95\%$ true positive rate on genuine hallucinations across multiple domains?

Sub-questions:

- What is the false rejection rate of existing verifiers (BERTScore, NLI, FActScore, GPT-4) on semantically valid transformed content?
- Do five derivation rules grounded in natural deduction provide sufficient coverage of naturally occurring transformations in practice?
- Does the framework generalize across domains (education, biomedical, legal)?

3. Theoretical Contributions

3.1 The Fidelity Assumption and Its Failure

Definition (Fidelity Assumption).

A verifier $V : \mathcal{L} \times \mathcal{KU} \rightarrow \{0, 1\}$ satisfies the Fidelity Assumption if there exists a similarity function $\text{sim} : \mathcal{L} \times \mathcal{L} \rightarrow [0, 1]$ and threshold $\tau \in (0, 1)$ such that:

$$V(c, \mathcal{KU}) = \mathbb{1} \left[\max_{ku \in \mathcal{KU}} \text{sim}(c, ku) > \tau \right]$$

Where:

- $\mathcal{L}$ denotes the space of natural language statements
- $\mathcal{KU}$ denotes the set of source knowledge units (factual statements extracted from retrieved docs)
- $ku \in \mathcal{KU}$ is an individual knowledge unit
- $c \in \mathcal{L}$ is a candidate claim generated by the LLM
- $\text{sim} : \mathcal{L} \times \mathcal{L} \rightarrow [0, 1]$ is a similarity function
- $\tau \in (0, 1)$ is a verification threshold

All existing verifiers satisfy this definition. BERTScore uses cosine similarity over BERT embeddings. NLI-based methods use entailment probability as a soft similarity measure. FActScore uses lexical overlap after atomic decomposition.

Theorem (Quantitative Impossibility of Surface-Form Verification).

Let V_{SF} be any threshold-based verifier satisfying the Fidelity Assumption. Define the transformation intensity $t \in [0, 1]$ such that $\text{sim}(\rho_t(ku), ku) = 1 - t$ (more intense transformations produce more surface-dissimilar outputs). Then for any threshold τ :

1. Every valid transformation with intensity $t > 1 - \tau$ is rejected regardless of semantic validity.
2. Under uniform transformation intensity $T \sim \text{Uniform}[0, 1]$: $\text{FRR}(V_{SF}) \geq \tau$, where FRR is the false rejection rate (the probability that a semantically valid adapted claim is incorrectly flagged as a hallucination).

3. The Precision-Adaptation Trade-off is unresolvable: $\text{FRR} + \text{FPR}_{\text{hallucination}} \geq 1$ for any τ .

Intuition. Raising τ catches more hallucinations but also rejects more valid transformations; lowering τ admits valid transformations but also admits hallucinations. No single threshold resolves this trade-off because validity and surface-form similarity are orthogonal properties. (Full proof in Appendix A.)

This theorem is domain-agnostic. It holds for any domain where valid transformations produce surface-form divergent outputs. The educational case is one instance; biomedical simplification, legal summarization, and code documentation are others.

3.2 The DERIVED Verification Category

We introduce a four-way verification taxonomy that resolves the impossibility by separating entailment-preserving transformations from surface-form matching:

Category	Definition	Action
GROUND	Claim c has direct lexical/semantic match with $\exists ku \in \mathcal{KU}$	Accept with citation
DERIVED	c is entailed by $\mathcal{KU}$ via a derivation rule $\rho \in \mathcal{R}$	Accept with derivation trace
GENERAL	c is common knowledge not requiring source grounding	Accept without citation
UNVERIFIED	Specific claim with no grounding or derivation	Flag for review

The DERIVED category does not exist in any prior verification framework.

3.3 Derivation Rules Grounded in Natural Deduction

The five derivation rules $\mathcal{R} = \{\rho_1, \dots, \rho_5\}$ are not arbitrary. The first four rules correspond exactly to inference rules of Gentzen’s (1935) natural deduction calculus restricted to operations that change surface form. The fifth is grounded in Gentner’s (1983) Structure-Mapping Theory, the most formal and empirically validated theory of analogy in cognitive science.

Master Definition (Semantic Entailment Preservation).

A transformation $\rho : \mathcal{L} \times \mathcal{KU} \rightarrow \mathcal{L}$ is *semantically preserving* with respect to knowledge unit ku if:

$$\llbracket ku \rrbracket \models \llbracket \rho(c, ku) \rrbracket$$

That is: in every world where ku is true, the derived claim $\rho(c, ku)$ is also true. This is entailment *from source to claim*, not containment: Instantiation $(\forall x.P(x) \models P(a))$ weakens the universal to a specific case, making the derived claim true in more worlds than the source — yet the source still entails the claim.

The five rules each instantiate this definition through a distinct logical operation:

Rule	Logical Basis	Formal Schema	Example
Substitution (ρ_{sub})	Leibniz’s Law / Congruence	$\llbracket F \rrbracket = \llbracket I \rrbracket \Rightarrow ku[F] \models ku[I]$	“ $O(\log n)$ ” $\rightarrow$ “grows very slowly”
Instantiation (ρ_{inst})	$\forall$ -Elimination	$\forall x.P(x) \models P(a)$	“ $O(\log n)$ ” $\rightarrow$ “~20 steps for $n = 10^6$ ”
Simplification (ρ_{simp})	$\wedge$ -Elimination (Weakening)	$P \wedge Q \models P$	“amortized expected $O(1)$ ” $\rightarrow$ “effectively constant time”
Implication (ρ_{imp})	Modus Ponens (Cut)	$P, P \rightarrow Q \models Q$	“ A is $O(n \log n)$, B is $O(n^2)$ ” $\rightarrow$ “ A is faster for large inputs”
Analogy (ρ_{ana})	Structure-Mapping (Gentner, 1983)	$P(a) \models P'(b)$ via structure-preserving map $\phi : a \mapsto b$	“Stack” $\rightarrow$ “pile of plates”

Theorem (Entailment Soundness of $\mathcal{R}$).

For $\rho \in \{\rho_{sub}, \rho_{inst}, \rho_{simp}, \rho_{imp}\}$, assuming a sound semantic oracle $\llbracket \cdot \rrbracket$ and correct rule-specific parsing:

$$\forall ku \in \mathcal{KU}, \forall c \in \mathcal{L} : \llbracket ku \rrbracket \models \llbracket \rho(c, ku) \rrbracket$$

Analogy (ρ_{ana}) is sound conditionally, assuming ϕ is a structure-preserving map between source and target domains.

Intuition. Each of the four deterministic rules is a tautology of first-order logic: Substitution holds by Leibniz congruence; Instantiation by universal instantiation; Simplification by conjunction elimination; Implication by modus ponens. Analogy introduces a domain-specific assumption that the structural mapping holds. (Full proof in Appendix A.)

Corollary (Compositional Closure). The DERIVED category is closed under finite composition: if c_1 is DERIVED from ku via ρ_i , and c_2 is DERIVED from c_1 via ρ_j , then c_2 is DERIVED from ku via $\rho_j \circ \rho_i$. This follows from transitivity of $\models$. (Proof in Appendix A.)

Completeness. We do not claim $\mathcal{R}$ enumerates all valid transformations, such completeness is provably unattainable for any finite rule set (infinite valid analogies exist; infinite instantiations exist). We claim $\mathcal{R}$ provides a *generative basis*: any finite composition of the five primitives produces a sound DERIVED claim. The coverage question, what fraction of naturally occurring transformations decompose into compositions of these primitives is empirical and measured in Experiment A1.

Two implementation notes. (i) ρ_{simp} is audience-dependent: dropping “amortized” from “ $O(1)$ amortized” is sound for a novice audience but semantically incorrect for an expert. (ii) Negation introduction ($\neg P$ derived from $P \rightarrow \perp$) is treated as a composite of ρ_{imp} and logical negation.

3.4 TAV Circumvents the Impossibility

Corollary (TAV Bypasses $\text{FRR} \geq \tau$ Barrier). TAV’s derivation rules are similarity-independent: $\text{derives}(c, \{ku_i\}, \rho)$ does not use $\text{sim}(c, ku)$. Therefore, the FRR of TAV is bounded by the *incompleteness* of $\mathcal{R}$, not by any threshold τ . For any valid transformation $\rho_t(ku)$ where ρ_t matches a rule $\rho \in \mathcal{R}$, TAV correctly classifies c as DERIVED regardless of surface-form divergence t . This is the formal justification for TAV’s existence.

3.5 Verification Function

$$\text{Verify}(c, \mathcal{KU}) = \begin{cases} \text{GROUNDED} & \text{if } \exists ku \in \mathcal{KU} : \text{match}(c, ku) \\ \text{DERIVED} & \text{if } \exists \{ku_j\} \subseteq \mathcal{KU}, \rho \in \mathcal{R} : \llbracket \{ku_j\} \rrbracket \models_{\rho} \llbracket c \rrbracket \\ \text{GENERAL} & \text{if } c \in \mathcal{G} \\ \text{UNVERIFIED} & \text{otherwise} \end{cases}$$

where $\mathcal{G}$ denotes the set of domain-general common-knowledge statements not requiring source citation.

Implementation:

- *Substitution and Instantiation:* Pattern matching with symbolic math (SymPy), fully deterministic.
- *Simplification and Analogy:* Prompted LLM classifier (confidence threshold = 0.7).
- *Implication:* Lightweight NLI (DeBERTa, entailment threshold = 0.8). Total overhead: < 50ms per response.

4. DERIVED-Bench: Multi-Domain Benchmark

The primary empirical contribution is DERIVED-Bench, the first benchmark designed specifically to evaluate verification systems on transformation-aware tasks.

4.1 Domain Selection

Three domains covering the breadth of the DERIVED Problem:

Domain	Source Corpus	Rationale
CS Education	Original CS curriculum corpus	Bloom’s taxonomy enables principled stratification; transformation types well-defined
Biomedical Simplification	PubMed abstracts + MedlinePlus patient summaries	Existing parallel corpus; extreme surface divergence; high stakes
Legal Summarization	EUR-Lex directives + plain-language government summaries	Formal-to-informal register shift; complex implication chains

4.2 Instance Structure

Each instance is a triple (ku, c, ℓ) where ku is the source knowledge unit, c is the candidate claim, and $\ell \in \{\text{GROUNDED}, \text{DERIVED-}\rho, \text{GENERAL}, \text{HALLUCINATION}\}$.

Label	Count per Domain	Total (3 Domains)
GROUNDED	100	300
DERIVED-Substitution	60	180
DERIVED-Instantiation	60	180
DERIVED-Simplification	60	180
DERIVED-Analogy	60	180
DERIVED-Implication	60	180
DERIVED-Composite (≥ 2 rules)	80	240
GENERAL	60	180
HALLUCINATION	100	300
Total	640	1,920

Composite instances (two or more rules in sequence) are the most theoretically important as they test Compositional Closure in practice. HALLUCINATION instances are adversarially constructed: each is paired with a DERIVED instance of the same surface form to maximize difficulty for surface-form verifiers.

4.3 Construction Protocol

Strategy B (LLM-generated, human-verified) for 80% of instances; **Strategy A (human-authored)** for 20% as quality anchors.

Strategy B prompt template:

“Given source statement: $[ku]$. Write: (1) a valid $[\rho]$ -type transformation — entailed by the source, uses different surface form; (2) a hallucination that resembles it but introduces an unsupported claim. Explain which rule applies and cite the source span.”

Annotation protocol. Three annotators per instance. Annotators label: (a) verification category, (b) rule type for DERIVED instances, (c) source span justification. Inter-annotator agreement reported at two levels.

Targets: Cohen’s $\kappa > 0.75$ for category label; $\kappa > 0.65$ for rule attribution (rule attribution is harder; acknowledging this is methodologically honest).

4.4 Cross-Domain Validation

Before full benchmark construction, 20–30 instances per domain will be manually annotated to verify that the five rules generalize. Specifically, PubMed $\rightarrow$ MedlinePlus transformations will be manually classified against $\mathcal{R}$. If rule coverage is below 70%, the rule set will be revised before construction.

5. Evaluation Design

Experiment A1: Benchmark Construction and Coverage Analysis

- Construct DERIVED-Bench across three domains (1,920 instances total).
- Measure rule coverage: what fraction of naturally occurring transformations in each domain decompose into the five rules.
- Target: $> 85\%$ coverage in CS domain; $> 75\%$ in biomedical and legal.
- Report decomposition distribution per domain.

Experiment A2: Baseline Failure Analysis (Validating the Impossibility Theorem)

Measure FRR of existing verifiers on DERIVED instances:

System	Description
BERTScore ($\tau = 0.85$)	Standard semantic similarity
NLI-DeBERTa	State-of-the-art NLI entailment
FactScore	Atomic fact decomposition + retrieval verification
GPT-4 prompted	Zero-shot LLM verifier (upper bound)

Hypothesis (from Theorem): all surface-form verifiers will show $\text{FRR} \geq 0.40$ on DERIVED instances, with highest FRR on Analogy and Composite subcategories.

Experiment A3: TAV Full Evaluation

Condition	Description
TAV-NoRules	Entity matching only (no derivation rules)
TAV-Symbolic	Substitution + Instantiation only
TAV-Full	All five rules

Metrics: Precision, Recall, F1 per verification category, FRR, TPR. Targets: TAV-Full $\text{FRR} < 10\%$, $\text{TPR} > 95\%$, $\text{F1}(\text{DERIVED}) > 0.85$, Overall Accuracy $> 90\%$.

Experiment A4: False Positive Reduction

- Dataset: 100 randomly sampled CLAG-adapted CS responses with known transformation types.
- Metric: FP reduction vs. BERTScore and NLI baselines.
- Target: $> 40\%$ FP reduction.

Experiment A5: Cross-Domain Generalization

Apply TAV-Full trained/configured on CS domain to biomedical and legal domains without retraining. Measure $\text{F1}(\text{DERIVED})$ drop across domains. This validates the domain-agnostic claim.

Statistical Analysis. McNemar’s test for accuracy comparisons. Paired t -test for FRR comparisons. Cohen’s κ for inter-annotator agreement.

6. Differentiation from Prior Work

System	What It Does	Why TAV Is Different
FActScore (Min et al., 2023)	Atomic fact decomposition + source verification	Assumes facts should match source; DERIVED category absent
VeriScore (Song et al., 2024)	Verifiable claim extraction + web search	Surface-form matching; no transformation rules
VeriCite (Qian et al., 2025)	Citation verification in RAG	Checks citation, not semantic transformation validity
FactLens (Mitra et al., 2025)	Benchmarks fact verification	Benchmark only; no derivation-aware category
GraphCheck (2025)	Graph-based post-generation fact checking	Graph-structural; no logical derivation rules

Key differentiator: TAV is the first framework to (a) formally prove the impossibility of surface-form verification for transformed content, (b) introduce the DERIVED category grounded in natural deduction, and (c) provide a multi-domain benchmark for evaluation.

7. Limitations and Honest Bounds

Claim	Status	Rationale
$\mathcal{R}$ is complete	Provably unattainable	Empirical coverage: $\mathcal{R}$ accounts for $X\%$ of observed valid transformations (Experiment A1)
TAV never produces false validations	Cannot prove	Soundness theorems are conditional on A1/A3; false validations occur when LLM classification is wrong
Rules generalize to all domains	Cannot prove	Validated empirically on 3 domains; generalization beyond these is future work
Analogy rule is fully deterministic	False	Analogy relies on LLM classification; determinism holds only for Substitution and Instantiation

PAPER 2: CLAG (Stateless Cognitive Load-Aware Generation for Educational RAG)

1. Problem Statement

Existing adaptive learning systems i.e., Intelligent Tutoring Systems (VanLehn, 2011), Bayesian Knowledge Tracing (Corbett & Anderson, 1995) universally assume persistent learner models: mastery tracking, error history, session analytics. For stateless RAG systems deployed on consumer hardware without login, database, or session tracking, these assumptions fail entirely.

Definition (Stateless Cognitive Adaptation). Given a single query q and retrieved context D , estimate the student’s cognitive level solely from (q, D) and produce a response adapted to that level without any session history, persistent user model, mastery score, error log, or prior interaction data.

The prevailing orthodoxy: the **Memory Dogma** holds that adaptation requires remembering the learner. CLAG’s core hypothesis is that a single query’s linguistic fingerprint contains sufficient signal for meaningful cognitive load estimation, and that the Memory Dogma can be relaxed for stateless deployment contexts.

2. Core Research Question

Can query-derived linguistic signals (Bloom’s level, specificity, complexity) provide sufficient information for principled cognitive load adaptation in stateless educational RAG, achieving $\geq 80\%$ agreement with expert cognitive-level assessment and measurable comprehension gain over fixed-level generation?

3. Theoretical Contributions

Notation:

- q denotes the student query
- D denotes the set of retrieved document chunks
- $\Theta \in \{1, \dots, 6\}$ denotes the student’s true Bloom’s level (latent)
- $\theta = (\Theta - 1)/5 \in [0, 1]$ is its normalized form
- $B(q) \in [0, 1]$ is the detected Bloom level (the primary signal)
- $S(q) \in [0, 1]$: Query specificity
- $L(q) \in [0, 1]$: Linguistic complexity

3.1 Mathematical Formulation

Query Demand:

$$QD(q) = B(q) + \text{clamp}\left(\beta_S \cdot (S(q) - \tfrac{1}{2}) + \beta_L \cdot (L(q) - \tfrac{1}{2}), -\delta_q, +\delta_q\right)$$

where $\beta_S = 0.3$ and $\beta_L = 0.2$ are sensitivity parameters, and $\delta_q = 0.10$ is the maximum auxiliary correction (half a Bloom level). The formula uses $B(q)$ as the base estimate; S and L contribute bounded corrections only.

Rationale. A naive linear formula $QD = w_1B + w_2S + w_3L$ allocates only 50% weight to Bloom’s level, allowing S and L to cause up to 2.5 Bloom levels of worst-case error. The clamped formulation guarantees $|QD - B| \leq 0.10$ always, eliminating this failure mode. Under perfect Bloom detection, $|QD - \theta| \leq \delta_q = 0.10$; under typical accuracy ($p = 0.833$), $|QD - \theta| \leq 0.30$ for 83.3% of queries.

Content Complexity:

$$CC(D) = 0.4 \cdot \text{ConceptDensity}(D) + 0.3 \cdot \text{Readability}(D) + 0.3 \cdot \text{AbstractionLevel}(D)$$

where each component is normalized to $[0, 1]$: ConceptDensity is the ratio of unique technical concepts to total tokens; Readability is inverse Flesch-Kincaid; AbstractionLevel is the proportion of abstract versus concrete content.

Adaptation Level:

$$AL(q, D) = \alpha \cdot QD(q) + (1 - \alpha) \cdot \text{clamp}(CC(D), QD(q) - \delta, QD(q) + \delta)$$

where $\alpha = 0.7$ (QD primacy weight) and $\delta = 0.2$ (content clamp radius). When $CC(D)$ falls outside the window $[QD - \delta, QD + \delta]$, it is clamped to the boundary, preventing retrieved content from overriding the query-derived estimate.

Output: 5-dimensional adaptation vector $\mathbf{a}(q) \in [0, 1]^5$ mapping to: vocabulary complexity, information density, example usage, scaffolding level, response length.

3.2 Key Theoretical Results

Theorem (ZPD Targeting Guarantee). Given parameters $\alpha = 0.7$, $\delta = 0.2$, $\delta_q = 0.1$:

$$|AL(q, D) - \theta| \leq |B(q) - \theta| + \delta_q + (1 - \alpha)\delta$$

Under perfect Bloom detection ($B(q) = \theta$): $|AL - \theta| \leq 0.16$, i.e., the system remains within one Bloom level of the true cognitive level, always. (Full proof in Appendix B.)

Theorem (Adaptation Monotonicity). For fixed D and $\Delta = QD(q_1) - QD(q_2) > 0$:

$$AL(q_1, D) - AL(q_2, D) \geq \alpha \cdot \Delta > 0$$

Higher query demand strictly produces higher adaptation level. (Full proof in Appendix B.)

Theorem (Information-Theoretic Viability). Under uniform prior $\Theta \sim \text{Uniform}\{1, \dots, 6\}$ and Bloom’s detection accuracy $p = 0.833$, by Fano’s inequality:

$$I(\Theta; B(Q)) \geq 1.547 \text{ bits}$$

representing 59.8% of the maximum entropy $H(\Theta) = \log_2 6 = 2.585$ bits. A single query captures more than half the available information about true cognitive level, providing the information-theoretic justification for the stateless paradigm. (Full proof in Appendix B.)

Honest bound on statistical advantage. CLAG does not claim statistical superiority over multi-query systems. A stateless estimator can only dominate a n -query system when feature informativeness is high ($\rho_{S,\theta} > 0.68$). CLAG’s advantages are practical: zero cold-start latency, zero session state, no privacy concerns, no database infrastructure. The theoretical framework presents CLAG as an efficient estimator that trades $O(1/\sqrt{n})$ convergence for practical deployability.

3.3 Theoretical Grounding

Theory	Citation	Connection to CLAG
Cognitive Load Theory	Sweller (1988)	Adaptation vector controls extraneous load
Bloom’s Revised Taxonomy	Anderson & Krathwohl (2001)	$B(q)$ maps queries to cognitive levels
Expertise Reversal Effect	Kalyuga (2007)	CLAG reduces scaffolding for high-QD queries
Zone of Proximal Development	Vygotsky (1978)	AL targets content slightly above current level
Desirable Difficulties	Bjork (1994)	Avoids over-simplification for advanced learners

3.4 Bloom’s Level Detection

Level	Patterns	Example
1: Remember	“what is”, “define”, “list”	“What is a stack?”
2: Understand	“explain”, “describe”, “how does...work”	“How does recursion work?”
3: Apply	“implement”, “write code”, “how to”	“How to implement quicksort?”
4: Analyze	“compare”, “why does”, “difference”	“Compare arrays vs linked lists”
5: Evaluate	“which is better”, “pros and cons”	“Should I use SQL or NoSQL?”
6: Create	“design”, “propose”, “improve”	“Design a cache system”

3.5 Differentiation from Prior Art

System	What It Does	Why CLAG Is Different
FEED (Samuel et al., 2025)	Bloom’s classification to evaluate LLM quality	CLAG uses Bloom’s as input signal for generation adaptation
AdaptLearn AI (Gupta et al., 2025)	Bidirectional personalization via user data	Stateful; requires persistent user profiles

System	What It Does	Why CLAG Is Different
Standard ITS (BKT, ECD)	Persistent mastery/error tracking	Requires session history; CLAG requires zero history
Adaptive-RAG (Jeong et al., 2024)	Complexity routing for retrieval	Routes retrieval; does not adapt response characteristics

4. Evaluation Design

Experiment C1: Bloom’s Detection Accuracy

- **Data:** 300 CS education queries (50 per Bloom’s level $\times$ 6 levels)
- **Annotators:** 2 CS educators, Cohen’s κ (target: > 0.75)
- **Conditions:** Random, Majority-class, CLAG-Patterns, CLAG-Full ($B + S + L$), Oracle-LLM (GPT-4)
- **Metrics:** Per-level accuracy, macro-F1, confusion matrix
- **Target:** CLAG-Full $> 80\%$ accuracy, macro-F1 > 0.75

This experiment also serves as the weight sensitivity analysis: varying β_S, β_L within $[0.1, 0.4]$ quantifies how robustly the QD formula performs across parameter choices.

Experiment C2: User Study for Adaptation Quality

- N : 60 CS students (between-subjects, 20 per condition)
- **Conditions:** Fixed-Beginner, Fixed-Advanced, CLAG-Adaptive
- **Task:** 10 CS questions across Bloom’s levels 1–5, pre/post comprehension test
- **Metrics:** Appropriateness (Likert 1–5), difficulty match (Likert 1–5), comprehension gain: $\frac{\text{post} - \text{pre}}{\text{max} - \text{pre}}$, engagement (Likert 1–5)
- **Analysis:** One-way ANOVA per metric, Bonferroni correction, effect sizes (Cohen’s d)
- **Target:** CLAG $>$ Fixed conditions at $p < 0.05$, $d > 0.5$

Experiment C3: Ablation Study

Condition	Description
No-CLAG	Fixed medium adaptation level
Bloom-Only	$QD = B(q)$ only
Bloom + Specificity	$QD = B + \beta_S \cdot S$
Full-CLAG	Complete formula

Purpose: quantify marginal contribution of each QD component and validate the $\delta_q = 0.10$ clamping choice.

Statistical Analysis. One-way ANOVA with Bonferroni correction. Effect sizes: Cohen’s d (target: > 0.5). Inter-annotator agreement: Cohen’s κ (target: > 0.75).

5. Limitations and Honest Bounds

Limitation	Status	Mitigation
CS domain only	Acknowledged	Architecture generalizes; domain tuning required for other subjects
User study $N = 60$	Sufficient for medium effect ($d > 0.5$); acknowledged	Power analysis: $N = 60$ gives $> 80\%$ power at $\alpha = 0.05$, $d = 0.5$
$\delta_q = 0.10$ is theoretically derived, not empirically optimized	Acknowledged	Sensitivity analysis in Experiment C3 tests $\delta_q \in \{0.05, 0.10, 0.15, 0.20\}$
Bloom’s detection fails for ambiguous queries	16.7% of queries misclassified at $p = 0.833$	Error concentrates at Analyze/Evaluate boundary; documented in confusion matrix
CLAG does not outperform multi-query systems statistically	Acknowledged	Advantage is practical (zero history), not statistical; framed correctly

References

- Anderson, L. W., & Krathwohl, D. R. (2001). *A taxonomy for learning, teaching, and assessing: A revision of Bloom’s taxonomy of educational objectives*. Longman.
- Asl, M. M., Nguyen, T. H., & Chen, Y. (2025). FAIR-RAG: Faithful adaptive iterative refinement for retrieval-augmented generation. *arXiv preprint arXiv:2510.22344*.
- Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185–205). MIT Press.
- Corbett, A. T., & Anderson, J. R. (1995). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Modeling and User-Adapted Interaction*, 4(4), 253–278.
- Fang, J., Meng, R., & Macdonald, C. (2025). KiRAG: Knowledge-intensive retrieval-augmented generation. *arXiv preprint arXiv:2502.18397*.
- Gentner, D. (1983). Structure-mapping: A theoretical framework for analogy. *Cognitive Science*, 7(2), 155–170.
- Gentzen, G. (1935). Untersuchungen über das logische Schließen. *Mathematische Zeitschrift*, 39, 176–210.
- Gupta, S., Kumar, R., & Sharma, A. (2025). AdaptLearn AI: Bidirectional personalization in educational systems. *IEEE Transactions on Learning Technologies*.
- Honovich, O., et al. (2022). TRUE: Re-evaluating factual consistency evaluation. *Proceedings of NAACL 2022*, 3905–3920.

- Jeong, S., et al. (2024). Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. *arXiv preprint arXiv:2403.14403*.
- Kalyuga, S. (2007). Expertise reversal effect and its implications for learner-tailored instruction. *Educational Psychology Review*, 19(4), 509–539.
- Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, 38(1), 23–31.
- Kasneci, E., et al. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, Article 102274.
- Min, S., et al. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *Proceedings of EMNLP 2023*, 12076–12100.
- Mitra, A., et al. (2025). FactLens: Benchmarking fine-grained fact verification. *Findings of ACL 2025*.
- Qian, Z., Wang, P., & Li, H. (2025). VeriCite: Citation verification in retrieval-augmented generation systems. *Proceedings of CIKM 2025*.
- Samuel, S., Mathew, J., & Kumar, A. (2025). FEEG: Framework for educational evaluation using generative AI. *Expert Systems with Applications*, 238, Article 121847.
- Saxena, A., et al. (2025). On citation generation and verification in retrieval-augmented generation. *arXiv preprint arXiv:2509.21557*.
- Song, S., et al. (2024). VeriScore: Evaluating the factuality of verifiable claims in long-form text generation. *arXiv preprint arXiv:2406.19276*.
- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285.
- Sweller, J., van Merriënboer, J. J. G., & Paas, F. (1998). Cognitive architecture and instructional design. *Educational Psychology Review*, 10(3), 251–296.
- VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4), 197–221.
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.
- Xu, F., Shi, W., & Choi, E. (2023). RECOMP: Improving retrieval-augmented LMs with compression and selective augmentation. *arXiv preprint arXiv:2310.04408*.
- Zhang, T., et al. (2020). BERTScore: Evaluating text generation with BERT. *Proceedings of ICLR 2020*.
- Zhang, Y., et al. (2024). LongCite: Enabling LLMs to generate fine-grained citations in long-context QA. *arXiv preprint arXiv:2409.02897*.

Appendix A: Full Proofs (TAV)

Proof (Quantitative Impossibility). If $t > 1 - \tau$, then;

$$\text{sim}(\rho_t(ku), ku) = 1 - t < \tau$$

Under the genericity condition (the derived claim is surface-dissimilar from all other $ku' \neq ku$),

$$\max_{ku'} \text{sim}(\rho_t(ku), ku') < \tau$$

so V_{SF} rejects. Under uniform $T \sim \text{Uniform}[0, 1]$:

$$\text{FRR} = P(T > 1 - \tau) = \tau$$

For the trade-off bound: any τ achieving $\text{FRR} < \tau_0$ requires $\tau < \tau_0$, which means hallucinations with $\text{sim} > \tau$ are accepted, so $\text{FPR} > 1 - \tau_0 \geq 1 - \text{FRR}$, giving $\text{FRR} + \text{FPR} \geq 1$. ■

Proof (Entailment Soundness). Each rule is a tautology of first-order logic under the stated assumptions: Substitution by Leibniz congruence (if $\llbracket F \rrbracket = \llbracket I \rrbracket$ then substituting I for F preserves denotation in any context by compositionality of $\llbracket \cdot \rrbracket$); Instantiation by the universal instantiation axiom ($\forall x. P(x) \vdash P(a)$ holds in every first-order model); Simplification by conjunction elimination ($P \wedge Q \vdash P$); Implication by modus ponens ($P, P \rightarrow Q \vdash Q$); Analogy by structural homomorphism under A3 (if ϕ preserves relevant predicates, $P(a)$ in source implies $P'(\phi(a))$ in target). ■

Proof (Compositional Closure). If $\llbracket ku \rrbracket \models \llbracket c_1 \rrbracket$ and $\llbracket c_1 \rrbracket \models \llbracket c_2 \rrbracket$, then by transitivity of $\models$: $\llbracket ku \rrbracket \models \llbracket c_2 \rrbracket$. ■

Appendix B: Full Proofs (CLAG)

Proof (ZPD Targeting Guarantee). The clamp in the AL formula satisfies:

$$|\text{clamp}(CC, QD - \delta, QD + \delta) - QD| \leq \delta$$

so;

$$|AL - QD| \leq (1 - \alpha)\delta$$

The clamped QD formula satisfies $|QD - B| \leq \delta_q$. By the triangle inequality:

$$|AL - \theta| \leq |AL - QD| + |QD - B| + |B - \theta| \leq (1 - \alpha)\delta + \delta_q + |B - \theta|$$

Substituting $\alpha = 0.7$, $\delta = 0.2$, $\delta_q = 0.1$:

$$|AL - \theta| \leq 0.06 + 0.10 + |B - \theta| = 0.16 + |B - \theta|$$

Under perfect detection $|B - \theta| = 0$, giving $|AL - \theta| \leq 0.16$. ■

Proof (Adaptation Monotonicity). Let $g(a) = \text{clamp}(c, a - \delta, a + \delta)$ for fixed $c = CC(D)$.

$$\begin{cases} g(a) = a - \delta(\text{increasing}) & \text{if } c < a - \delta \\ g(a) = c(\text{constant}) & \text{if } a - \delta \leq c \leq a + \delta \\ g(a) = a + \delta(\text{increasing}) & \text{if } c > a + \delta \end{cases}$$

So g is non-decreasing. Then $AL_1 - AL_2 = \alpha(QD_1 - QD_2) + (1 - \alpha)(g(QD_1) - g(QD_2)) \geq \alpha\Delta + 0 = \alpha\Delta > 0$. ■

Proof (Information-Theoretic Viability). By Fano's inequality, for any estimator with error probability $P_e = 1 - p = 0.167$ over $K = 6$ classes:

$$H(\Theta|B(Q)) \leq h_b(P_e) + P_e \log_2(K - 1)$$

Computing: $h_b(0.167) = 0.6506$ bits, $0.167 \times \log_2(5) = 0.3878$ bits. So $H(\Theta|B(Q)) \leq 1.038$ bits. Under uniform prior $H(\Theta) = \log_2 6 = 2.585$ bits:

$$I(\Theta; B(Q)) = H(\Theta) - H(\Theta|B(Q)) \geq 2.585 - 1.038 = 1.547$$

bits. ■